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NPS/002/020 – Technical Specification for 11 & 20kV Power Cables

1. Purpose

The purpose of this document is to detail CE Electric UK's requirements for underground cable for use on distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

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NPS/002/020	2	Specification for 11 & 20kV Power Cables

2. Scope

This specification details the type, size and requirements for 11kV and 20kV power cables for use on CE Electric UK distribution network.

3. Technical Requirements

3.1 Conditions of Installation.

Cables specified in this document shall be installed in accordance with the CE Electric UK policy for the installation of cables: DSS/026. Cables will be laid "direct" into the ground, pulled into buried ducts, open trenches or installed in air. Cables may also be installed directly using trenchless excavation techniques. During storage and after installation cables can be expected to be subjected to the full range of climatic conditions encountered in the UK.

Cable laid "direct", in ducts or trenches may be surrounded by ground water for most of their operating lives. Cables installed above ground will be supported by means of cleats either vertically or horizontally and may be exposed to direct sunlight for significant periods. Where installed on wood poles these cables may come into contact with a pole preservation medium such as creosote.

Accessories required to joint the cable may be cold applied or require the application of heat.

3.2 Conditions of Operation

Power cables purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/909. The following are general conditions of operation:

- Nominal system voltages: 20kV and 11kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 6%.
- Nominal system frequency: 50Hz.
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

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3.3 11kV and 20kV Distribution Cables - Extruded Insulation

11kV and 20kV distribution cables of single core extruded insulation design will require to be supplied in both "true single core" and in "triplex" configurations – conductors' sizes as specified in Appendix 1. Cables shall comply in all respects with the requirements of BS 7870: 4.10 and meet any additional requirements specified in ENATS 09-17.

3.3.1 Conductors

The cable conductor's shall be circular solid aluminium or, circular compacted, stranded copper in accordance with BS EN 60228. The range of conductor cross sections required is as detailed in Appendix 1.

3.3.2 Insulation and Screens

The conductor screen, insulation and insulation screen shall be applied as a continuous single pass triple extrusion free of factory repairs. The cross linking process will be completely "Dry Cured" and no water will be used during this process. Maximum permissible shrinkage shall be 2%.

For designs with XLPE insulation, the insulation screen shall be of the "fully bonded" type. Insulation thickness for 11kV cables shall comply with requirements of BS 7870: 4.10 and any additional requirements of ENATS 09:17. Insulation thickness for 20kV cables shall comply with the requirements of BS6622 - 12.7/22(24)kV cables.

All cables shall comply with the concentricity and circularity requirements of BS 7870: 4.10 and meet any additional requirements specified in ENATS 09-17.

3.3.3 Metallic Screen and Moisture Blocking

All cables shall have minimum 35mm² copper wire screens and shall be blocked against longitudinal moisture transmission around the screen wires. Moisture blocking tape shall be provided under and over the copper wires. Moisture content of moisture blocking tape will be less than 50,000 ppm.

Cables shall pass the moisture penetration test outlined in IEC 60840: 2004.

3.3.4 Oversheath

The oversheath (coloured red for 11kV cables and black for 20kV cables) - shall consist of an extruded layer of MDPE complying with the requirements of BS 6234 and the relevant sections of BS 7870 part 4. , Maximum permissible shrinkage shall be 2%. The compound shall have a density within the range quoted in ENATS 09-17.

An outer graphite coating is not required.

3.3.5 Long Term Ageing

Cable designs must have successfully undergone CIGRE long term ageing tests as per proposed amendment 3, clause 5.4.15 to CENELEC HD 605. See also section 3.3 and clause 4.5.

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4. References

4.1 External Documentation

The cables shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

4.1.1 British Standards

Reference	Title
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS EN 60228	Conductors in insulated cables
BS 7870	LV and MV Polymeric insulated cables for use by distribution and generation utilities
BS 6622	Electric Cables – Armoured cables 6.6/33kV.....

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

4.1.2 Energy Networks Association Technical Specifications

Reference	Title
ENATS 09 - 17	Single core cable for use in substations having extruded insulation and rated voltages of 6350/11000 volt and 19000/33000 volt

4.2 Internal documentation

Reference	Title
IMP/001/909	CoP for Distribution System Parameter
DSS/026	Policy for the Installation of Distribution Power Cables

5 Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	05/10/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	21/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	27/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system.

		Sign	Date
S Johnson	DBD Administrator	S Johnson	02/11/2009

6.5 Approval

Approval is given for the content of this document.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	27/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document.

		Sign	Date
C Morgan	Head of Procurement	C Morgan	27/10/2009

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Appendix 1 – Cable Types Utilised

11kV Cables

Single core

- 95mm² Circular Solid Aluminium Conductor
- 185mm² Circular Solid Aluminium Conductor
- 300mm² Circular Solid Aluminium Conductor
- 300mm² Circular Stranded Copper Conductor
- 400mm² Circular Stranded Copper Conductor
- 630mm² Circular Stranded Copper Conductor

Triplex Formation

- 95mm² Circular Solid Aluminium Conductor
- 185mm² Circular Solid Aluminium Conductor
- 300mm² Circular Solid Aluminium Conductor
- 300mm² Circular Stranded Copper Conductor
- 400mm² Circular Stranded Copper Conductor
- 630mm² Circular Stranded Copper Conductor

20kV Cables

Single Core

- 95mm² Circular Solid Aluminium Conductor
- 185mm² Circular Solid Aluminium Conductor
- 400mm² Circular Stranded Copper Conductor

Triplex Formation

- 95mm² Circular Solid Aluminium Conductor
- 185mm² Circular Solid Aluminium Conductor
- 400mm² Circular Stranded Copper Conductor

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Appendix 2 – Addendum to Supplier Requirements

For each power cable offered the Tenderer shall provide the following cable rating data: -

- Cable rating laid direct (Amps)
- Cable rating laid in a 150mm internal diameter duct (Amps)
- Cable rating in air (Amps)
- Maximum dc resistance per phase conductor at 20⁰C (Ohms/km)
- Maximum ac resistance per phase conductor at maximum conductor temperature (Ohms/km)
- Star reactance at 50Hz (Ohms/km)
- Star capacitance at 50Hz (Ohms/km)
- Charging current per phase at normal voltage and frequency (mA/m)
- Zero sequence impedance $R_0 + jX_0$ (Ohms/km)
- Minimum dynamic bending radius (mm) (In Triplex and single core formations)
- Minimum static bending radius (mm)
- Recommended pulling method and maximum pulling tension (kgF)

The following assumptions shall be made when quoting ratings: -

Single core cables are laid in touching trefoil.

Cover to top of 11kV cables being 600mm

Cover to top of 20kV cables being 800mm

Ground Thermal Resistivity = 1.2 K m/W

Ground temperature = 15⁰C

Air temperature = 25⁰C

The oversheath of all cables shall be marked in accordance with the requirements of the relevant Standard specified in this document.

All cables shall be metre marked throughout the length of the cable and the start and end values shall be marked on the drum label.

Cables shall be marked with some form of reference or batch number that can be used to ensure batch trace ability of materials and manufacturing facilities used in the construction of the cable.

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Appendix 3 – Logistical Requirements

Drums used for 11 and 20kV cables shall have a maximum width of 1200mm and a maximum weight of 2500kg. All cable drums shall be returnable and the tenderer shall arrange to collect empty drums from the company's normal delivery locations.

Tenderers shall state at the time of tender their proposed cable drum sizes and weights for each cable type offered.

All cable drums shall be marked in accordance with the relevant cable Specification or Standard. The drum label shall also contain

- (a) Customers catalogue number
- (b) Name of manufacturer
- (c) Supplied length
- (d) Rated voltage
- (e) Number of cores
- (f) Size of conductor
- (g) Type of conductor material ("Cu" or "Al")
- (h) Abbreviated description of cable construction
- (i) Gross and nett weights
- (j) Direction of rolling drum
- (k) The metre marking starts and end values
- (l) The unique reference number

Cable drums may be stored for long period's outdoors. All drum labels shall remain legible and durable under these conditions.

The ends of all cables shall be effectively sealed against the ingress of moisture by a method appropriate to the cable type. Tenderers shall detail at the time of tender their proposed sealing arrangement for each cable type offered.

The cable end projecting from the drum shall be protected from damage during transit, storage and handling on site.

The cable on the drum shall not be susceptible to damage during transit, storage and handling on site. Tenderers shall state at the time of tender their proposed method of protection for each cable

Each delivery length of cable shall be allocated a unique reference number. This number shall appear on the factory test sheet covering the cable length, shall be clearly marked on the drum on which the length is delivered and shall be referred to on all invoices and advice notes.

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

11 and 20 kV cables required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of cable offered.

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	Clause / Requirements	Conformance Code	Remarks / Comments	
Conductor	BS EN 60228			
Conductor Screen Resistivity	ENATS 09:17 2.3.1.1 BS 7870 4:10 8.1.1 BS 6622 19.2			
Conductor Screen, Insulation & Insulation Screen: - Concentricity & Circularity	Continuous single pass triple extrusion (free from factory repairs)			
	Dry Cured Cross Linking			
	BS 7870 4:10 7.4 ENATS 09:17 2.1.4 BS 6622 18.7			
Insulation Thickness -11kV -20kV	BS 7870 4:10 4.2.3 ENATS 09:17 2.1.3			
	BS 6622 7 (Table 1)			
Insulation Test	ENATS 09:17 2.3.1.2 (Table 3) BS 7870 4:10 8.1.2 (Table 6)			
Insulation Shrinkage	Max 2% (refer to table 6: ENATS 09:17 & BS 7870 4:10 for test methods).			
Insulation Screen: - Resistivity	ENATS 09:17 2.3.1.3 BS 7870 4:10 8.1.3 BS 6622 19.4			
	Fully Bonded			
Screen	Min 35mm ² Cu			
Longitudinal - Tapes Moisture - Moisture Blocking: Tapes : Content	Under and Over Metallic Screen Wires			
	≤ 50 000 PPM			
Moisture Penetration Test	IEC 60840			

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		Clause / Requirements	Conformance Code	Remarks / Comments
Oversheaths : - 11kV Cables - 20kV Cables - Tests		Extruded MDPE to BS 6234, BS 7870 Part 4 & Density as ENATS 09:17		
		Coloured Red		
		Coloured Black		
		ENATS 09:17 2.3.1.4 BS 7870 4:10 8.1.4		
		Max 2% (refer to table 7: BS 7870 4:10 for test methods).		

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		Clause / Requirements	Conformance Code	Remarks / Comments
	Compatibility Test	ENATS 09:17 2.3.1.5 BS 7870 4:10 8.1.5 BS 6622 19.3		
	Partial Discharge Test	ENATS 09:17 2.3.2.1 BS 7870 4:10 8.2.1 BS 6622 17.5		
	Bending Test	ENATS 09:17 2.3.2.2 BS 7870 4:10 8.2.2 BS 6622 20.3		
	Power Factor Test	ENATS 09:17 2.3.2.3 BS 7870 4:10 8.2.3 / 8.2.4 BS 6622 20.4 / 20.5		
	Heat Cycling Test	ENATS 09:17 2.3.2.4 BS 7870 4:10 8.2.5 BS 6622 20.6		
	Impulse Withstand Test	ENATS 09:17 2.3.2.5 BS 7870 4:10 8.2.6 BS 6622 20.7		
	4 Hr Voltage Test	ENATS 09:17 2.3.2.6 BS 7870 4:10 8.2.7 BS 6622 20.8		
	Screen Adherence @ SC	ENATS 09:17 2.3.2.7 BS 7870 4:10 8.4 BS 6622 20.9		
	Cold Bend Test	ENATS 09:17 2.3.2.8 BS 7870 4:10 8.1.6		
	Long Term Ageing	CIGRE / CENELEC HD 605		